KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2021
Department of Computer Science and Engineering
CSE 4111
Machine Learning

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What do you mean by "Machine" and "Learning"? What are their individual features? In what prospects both the terms combine in Machine Learning (ML) subject? (07)
   b) "Training appropriate neural network architecture with appropriate training set is necessary to develop an appropriate model for a given task" – Justify the statement for a logic gate function case. (08)
   c) Draw the minimal neural architectures for the following three input (A, B and C) logic gate operations. (12)
      i) $NN1 = (AB)'C$ ii) $NN2 = ABC'$
      iii) $NN3 = (A + B)'C$ iv) $NN4 = (A + B)C'$
   d) Differentiate learning and generalization in the context of machine learning. Which one is more desirable and why? What are the measuring techniques of those? (08)

2. a) Why is a single neuron unable to solve XOR problem? Solve XOR problem with Radial Basis Function (RBF) network considering cluster centers (0, 1) and (1, 0) with appropriate radius of each center. (08)
   b) Write down the roles of activation function in artificial neural network? Compare two different forms of sigmoidal functions through graphical representations of Input vs Output and derivatives. (08)
   c) What is the naming significance of "Back Propagation (BP)" algorithm? What is the effect of learning rate in BP? How momentum parameter speedup training in BP? (07)
   d) What is the goal of neural network ensemble construction? Explain an ensemble method for enough data and an ensemble method for limited data. (12)

3. a) Why non-deterministic operation is preferable over deterministic operation in population-based optimization algorithms? How non-deterministic feature assures in Genetic Algorithm (GA), Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO)? (12)
   b) Explain velocity and position updates of particles in PSO mentioning corresponding equations in continuous function space. How the terms may transfer to solve Traveling Salesman Problem (TSP)? (09)
   c) Why both exploration and exploitation operations are necessary in an algorithm for better performance? (06)
   d) How individual parameters of PSO velocity update equation influence exploration and exploitation? (08)

4. a) What do you understand by the search terms Global Search, Local Search and Blind Search? Compare GA with a prominent Swarm Intelligence (SI) method on the basis of the search methods. (13)
   b) Explain the working principle of GA with a flowchart. Briefly explain importance of different operators and parameters in GA. (08)
   c) Suppose two TSP solutions are 1 – 2 – 4 – 3 – 6 – 5 and 1 – 5 – 3 – 4 – 2 – 6 where numeric values are city indexes. Generate an offspring from the solutions using Enhanced Edge Recombination algorithm. (08)
   d) Explain the fundamental differences between ACO and PSO to establish interaction among individuals. (06)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is 'K' in the K Nearest Neighbor (KNN) algorithm and why is the odd value of 'K' preferred over even values in the KNN algorithm? (05)

   b) We can easily find K Nearest Neighbors using Euclidean distance and then classify an instance. Then, why do we need K Dimensional tree (K-D Tree) for KNN? (05)

   c) Suppose, a satellite image can be segmented using four categories: water body, grass land, bare land and semi grassy area. Choose a suitable algorithm to segment the image and explain the process. (05)

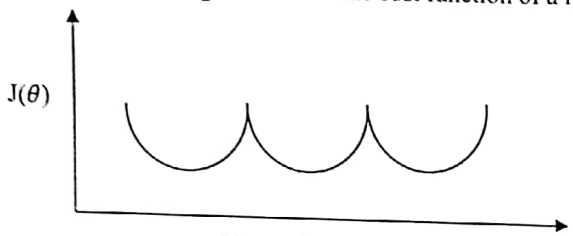
   d) Consider the following cost function and hypothesis for linear regression. Derive the normal equation, $\theta = (X^T X)^{-1} X^T Y$ to get the values of vector θ. (15)

$$j(\theta) = \frac{1}{2m}\sum_{i=1}^{m}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)^2$$

$$h_\theta\left(x^{(i)}\right) = \theta_0 + \theta_1 x_1^{(i)} + \cdots + \theta_n x_n^{(i)}$$

   How do you calculate θ when $X^T X$ is non-invertible?

   e) Consider the following outcome of the cost function of a linear regression model. (05)



   Is there any issue with the outcome? Explain the issue and solution (if any).

6. a) The cost function of logistic regression is given below: (15)

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log\left(h_\theta\left(x^{(i)}\right)\right) + \left(1 - y^{(i)}\right)\log\left(1 - h_\theta\left(x^{(i)}\right)\right)\right]$$

   Prove that, $\frac{\partial}{\partial\theta_j}J(\theta) = \frac{1}{m}\sum_{i=1}^{m}\left[h_\theta\left(x^{(i)}\right) - y^{(i)}\right]x_j^{(i)}$.

   b) Is there any similarity between a logistic regression model and a single neuron of a neural network (with sigmoid activation function)? Explain (if any). (07)

   c) What do you mean by overfitting and underfitting? Write down a way to minimize both the issues. (07)

   d) Suppose, one of your friends is planning to build a Machine Learning (ML) model to predict the diseases of many patients from various symptoms. He wants to sell the product to several hospitals. He has only 400 instances of data from 300 different patients. The dataset has both categorical and continuous features. There are only 5 diseases for prediction. Which ML model would you suggest to your friend and why? (06)

7. a) Consider the following dataset where attribute 'Salary' represents class label. (14)

Department	Age	Status	Salary
Sales	25..30	Junior	Low
Sales	31..35	Senior	Medium
Marketing	31..35	Senior	High
Admin	25..30	Junior	Medium
Admin	31..35	Senior	High
Admin	36..40	Senior	High
Marketing	36..40	Senior	High
Sales	25..30	Junior	Low
Marketing	36..40	Senior	Medium
Marketing	25..30	Junior	Low

   Find the root attribute of the Decision Tree (created from the dataset) using Gain Ratio as the attribute selection measure.

b) Consider the following problem. (14)

Minimize $f(x, y) = (x - 1)^2 + (y - 3)^2$

Subject to $x + y \leq 2, y \geq x$

Use the following formula for the minimization problem and find the minimum values for x and y.

Lagrangian, $L = f(x) + \sum_{i=1}^{p} \lambda_i \cdot h_i(x)$.

KKT constraints are as follows: $\frac{\partial L}{\partial x_i} = 0, i = 1,2, \cdots\cdots, d.$

$\lambda_i \geq 0, i = 1,2, \cdots\cdots, p.$

$h_i(x) \leq 0, i = 1,2, \cdots\cdots, p.$

$\lambda_i \cdot h_i(x) = 0, i = 1,2, \cdots\cdots, p.$

c) The learning task in Support Vector Machine (SVM) can be formulated as the following constrained optimization problem. (07)

Minimize $\mu(W, b)$

Subject to $y_i(W \cdot x_i + b) \geq 1, i = 1,2,3, \cdots\cdots, n.$

Write down the Lagrangian and KKT constraints for this problem.

8. a) Consider the case of a binary classification with a training data of 6 tuples as shown in the following table. (12)

X_1	X_2	Y	λ_i
0.38	0.48	+	65.52
0.59	0.71	−	65.52
0.82	0.31	−	65.52
0.75	0.90	−	0
0.19	0.59	+	0
0.41	0.35	+	0

The lagrange multipliers λ_i for each training tuple are solved using quadratic programming and the first three tuples are support vectors. Determine W and b to find the Maximum Margin Hyperplane of linear SVM.

b) Briefly explain the curse of dimensionality problem. (06)

c) Consider the following dataset with 2 dimensions. (17)

Dimension 1	Dimension 2
2.5	2.4
2.2	2.9
1.9	3.0
2.7	2.5
2.0	1.6
2.1	1.1
1.0	1.8
1.9	2.7

Using Principal Component Analysis (PCA), reduce the dataset to one dimension. Explain each step with necessary calculations.